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Deep Reinforcement Learning based approach for Traffic Signal Control

Kővári Bálint[a,*], Tettamanti Tamás[a], Bécsi Tamás[a]

[a]*Department of Control for Transportation and Vehicle Systems, Budapest University of Technology and Economics, Műegyetem rkp. 3, 1111, Budapest, Hungary*

Abstract

The paper introduces a novel approach to the classical adaptive traffic signal control (TSC) problem. Instead of the traditional optimization or simple rule-based approach, Artificial Intelligence is applied. Reinforcement Learning is a spectacularly evolving realm of Machine Learning which owns the key features such as generalization, scalability, real-time applicability for solving the traffic signal control problem. Nevertheless, the researchers' responsibilities become more serious regarding the formulation of state representation and the rewarding system. These Reinforcement Learning features are also the most fascinating and controversial virtues since the utilized abstractions decide whether the algorithm solves the problem or not. This paper proposes a new interpretation for the feature-based state representation and rewarding concept that makes the TSC problem highly generalizable and has scaling potential. The proposed method's feasibility is demonstrated via a simulation study using a high-fidelity microscopic traffic simulator. The results justify that the Deep Reinforcement Learning based approach is a real candidate for real-time traffic light control.

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Keywords: Deep Reinforcement Learning; Traffic Light; Traffic Signal Control; Policy Gradient Algorithm; SUMO; Traffic Simulation

1. Introduction

The presence of Deep Reinforcement Learning (DRL) based approaches in the diverse realm of road transportation is growing dramatically Farazi et al. (2020). Most aspects of autonomous vehicle control, such as motion planning, highway driving, etc., are intensely researched with DRL techniques Aradi (2020). The same goes for the competitive problem of route optimization, ranging from the fundamental traveling salesman problem Bello et al. (2016) to more complex multi-vehicle routing problems Nazari et al. (2018). Adaptive Traffic Signal Control (TSC) also at the center of interest. This trend can last because DRL can provide a natural and robust framework for solving complex sequen-

* Corresponding author.
E-mail address: kovari.balint@kjk.bme.hu

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tial decision-making problems. Moreover, thanks to the deep learning component, it can provide real-time solutions in challenging domains which computationally overwhelms classic methods. Traffic light control is a permanent engineering topic as traffic problems emerge worldwide, and autonomous vehicles will likely generate more mobility than before (Kisgyörgy and Szele, 2018). Traffic-responsive signal control is present in traffic engineering practice for several decades to tackle traffic congestion. With widespread traffic sensors (loop detectors, earth-magnetic detectors, cameras, etc.) it is used increasingly intensively (Papageorgiou, 2004).

Deep Reinforcement Learning based control has been applied for traffic light control in a limited manner. Most of the authors utilize Value-based algorithms in Single-Agent solutions such as Deep Q-Learning (Guo and Harmati, 2020; Touhbi et al., 2017) and its on-policy variance SARSA (El-Tantawy et al., 2014; Wen et al., 2007) which, compared to Policy-based methods, can not guarantee convergence at least to some local optima. Hence, these methods are more exposed to struggle in convergence during training, which, as a result, can shrink performance. Moreover, most of the used reward strategies utilize low-level features like queue length (Muresan et al., 2019), waiting time Gao et al. (2017) or no traffic dependent features at all (Thorpe and Anderson, 1996), making credit assignment and generalization harder.

The paper's main contribution is twofold. On the one hand, it is justified that Deep Reinforcement Learning can be effectively used for traffic-actuated signal control problem. On the other hand, this paper proposes a new interpretation of the composed feature-based state representation that enables the utilization of a new rewarding concept. Considering the rewarding systems' unique role in training procedures, it can firmly enhance the TSC problem's generalization, which is crucial since generalization is a critical component in RL agent's adaptation capabilities for unseen scenarios. It is also worth mention that in this paper, the Policy Gradient algorithm is utilized, which has guaranteed convergence. Thanks to this feature, it has the potential to reach better performance, compared to that Value-based approaches, which are utilized mostly for the TSC problem in the past (Haydari and Yilmaz, 2020).

The paper is structured as follows. In Section 2, the applied methodology for reinforcement learning is discussed. Section 3 introduces the applied working environment used for machine learning and traffic simulation. Section 4 demonstrates the results based on realistic traffic simulation. The paper ends up with a short conclusion.

2. Methodology

2.1. Reinforcement Learning

Reinforcement Learning (RL) is a currently heavily researched realm of Machine Learning (ML) thanks to its tremendous success in boardgames (Silver et al., 2017), robotics (Lillicrap et al., 2015), videogames (Mnih et al., 2015). Compared to Supervised Learning (SL), RL utilizes a different concept of learning. An SL agent seeks to map the input x to the output y based on training samples. Hence the primary concerns are the insatiable demand for training data, which can not be provided in several control problems. In the meantime, RL generates its training data by interacting with the environment in an online manner. The interaction between an RL training loop's entities is formulated as a Markov Decision Process (MDP) $\{S, A, T, R\}$, where the agent seeks to maximize the cumulative reward by developing a policy. The cumulated reward is formulated as follows:

$$G = \sum_{t=1}^{T} \gamma^t r_t. \tag{1}$$

Where γ is the discount factor that determines the dependence of past actions on future rewards. An interaction looks as follows: The agent receives state s_i that represents the environment. The agent chooses an action a_i based on a policy π_i that execution triggers a state transition $P(s_i, a_i|s_{i+1})$ in the environment. The quality of the selected action a_i in the given time step is described to the agent by the environment through a scalar feedback value called reward. In the RL concept, the reward signal can be interpreted as the only compass that an agent has to develop the optimal policy.

2.2. Policy Gradient algorithm

For solving the introduced control task, the Policy Gradient algorithm is implemented, which can be categorized into the family of model-free Policy-based RL algorithms. An agent's training is basically about tuning the utilized function approximator, which is a Neural Network in our case. If the training is successful, then the function approximator predicts values or probabilities from which the policy can be derived directly or indirectly. Compared to Value-based methods, the Neural Network trained via the Policy Gradient algorithm predicts the choice probabilities of the available actions in every state, which can be interpreted as direct suggestions of choosing one or another. More specifically the output of the Policy Gradient algorithm is a probability distribution, which is defined by the function approximator's θ parameters. Consequently the parameters of the function approximator have to be tuned precisely to maximizes the agent's performance indicator function which is identical to the reward signal acquired from the environment $J(\theta) = J(\pi_\theta)$. The mentioned function looks as follows:

$$J_{\pi_\theta} = \mathbb{E}\Big[\sum_{t=1}^{\tau} \gamma^t r_t\Big] \tag{2}$$

The underlying idea behind the equation above is to find local extreme values in the direction of the greatest increase of the reward signal. Considering Williams (1992), and Sutton et al. (2000), the updated rule of the Policy Gradient algorithm is formulated as follows:

$$\theta \leftarrow \theta + \alpha \nabla \log \pi_\theta(s_t, a_t) \sum_{t=1}^{\tau} \gamma^t r_t \tag{3}$$

and steps of its operation are:

1. The training starts from the initial state s_0 of the environment, while the function approximator's θ parameters are generated according to the chosen initial scheme.
2. After the initialization of the training loop's two entities, they interact until a terminal state encounters.
3. The interactions $\{S, A, R\}$ are saved to the episode history, and the discounted rewards are calculated via the chosen γ discount factor.
4. Finally, the gradients are calculated and added to the gradient buffer, which updates the function approximator's weights if the updated frequency encounters.

$\pi_\theta(s_t, a_t)$ stands for the probability of choosing action a in, state s and the α parameter is the learning rate, which defines how firmly a gradient can change the function approximator's parameters at once.

3. Environment

As mentioned, the chosen problem is the Traffic Signal Control (TSC) of a simple intersection, where the vehicles can arrive from all four directions on one lane each. The left turns in the intersection are omitted for simplicity. Hence the vehicles are only allowed to turn right or go straight out from the network. The training environment is modeled in SUMO (Simulation of Urban Mobility) (Lopez et al., 2018). SUMO is a perfect tool for such an endeavor since it can utilize both real transportation networks and artificially created ones with tremendous options to customize and scale. On top of all that, it has an excellent Python interface called TraCI, which is crucial from the aspect of Deep Reinforcement Learning, since most of the development tools and abstractions like Tensorflow, PyTorch, and Keras are implemented in Python. Figure 1 shows how the SUMO is integrated into the introduced RL training loop. The described infrastructure is formulated in SUMO and controlled from the Python training environment via TraCI.

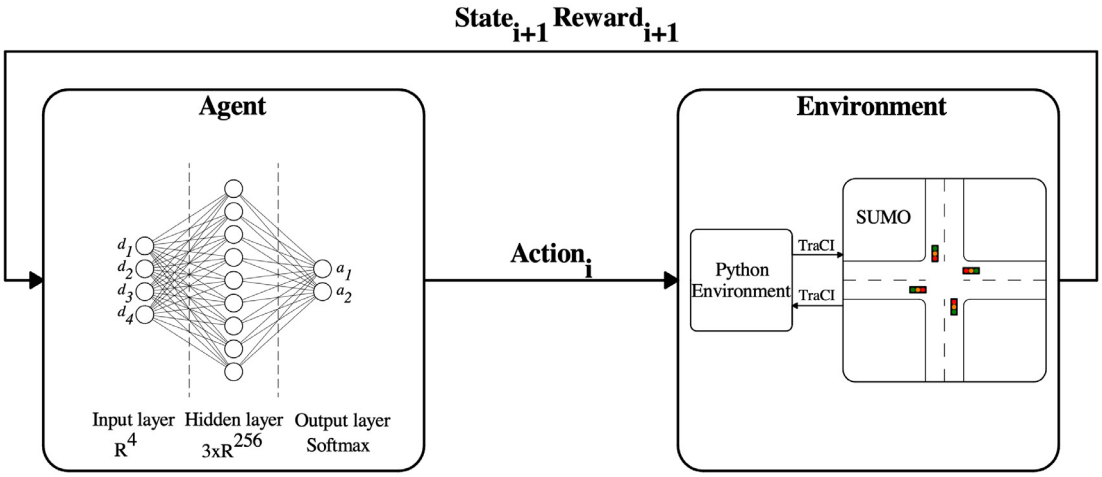

Fig. 1. The Reinforcement Learning training loop

The Python training loop is designed according to the OpenAI gym standards. Hence the decisions of the agent are channeled into TraCI commands by the Python environment. Then the TraCI commands trigger changes in the SUMO infrastructure. The change's measures are acquired by TraCI and formulated appropriately by the Python environment to represent the infrastructure's state to the agent.

Deep Reinforcement Learning's main advantages and capabilities are real-time applicability, generalization, and scalability of the solution, which means that the trained Neural Network can manage unseen scenarios in the right fashion. In establishing high-level generalization and scalability, most of the attention goes to the algorithms. Still, the state vector's importance that represents the environment to the agent also has to be valued appropriately since this is the part of DRL where the algorithm can assign credit, or in other words, figure out how each piece of information affects one or another. Consequently, the composition of the control task's abstraction is the key to a scaleable and generalizable solution. Hence state representation is one of the most crucial aspects since a state descriptor's conciseness directly influences the training procedure's success. Moreover, the formulation of the state descriptor is exclusively the researcher's duty. Thus the result depends on how well the researcher knows the abstraction of the control problem. Nevertheless, the same consideration goes for the formulation of the reward signal in DRL because this is the only compass that the agent has in finding the optimal behavior. In this paper, our angle is to compose a state vector and a rewarding scheme for the TSC problem that can be easily generalized and has the potential of scaling.

3.1. State representation

Image like state representations is widely spread in the realm of traffic signal control problems. The majority of the researchers favor Discrete Traffic Signal Encoding (DTSE), where all the lanes in the intersection are divided into cells starting from the lane's end. The average length of the vehicles determines the cell size. The cells are filled with different types of information, such as the pace, acceleration, vehicle position, and signal phase. This matrix can be

fed into a Convolution Neural Network (CNN) with n channels, where cells are interpreted as pixels, and n defines the number of different types of information represented in the given cells. Other image-like representations are raw RGB images that can include roadside information, which can be advantageous in several scenarios. Another common approach in state representation is feature-based value vectors. In this case, every lane of a road is represented with a set of values such as phase duration, average speed, cumulative waiting time, phase cycle, queue length. This concept's main advantage is that the utilized information can be easily gathered from the intersection with loop detectors or road sensors.

In our case, a feature-based approach is utilized, which solely contains one value for each lane. This value is the number of vehicles currently occupying the particular lane, which is not the same as the queue length. The queue length is the feature mostly utilized in the state representation of the TSC problem. Still, it is hard to estimate it correctly, and in some sense, it carries the information about the lane's traffic load in a delayed manner compared to this approach. The values are transformed into [1,0] interval to mitigate the effect of both vanishing and exploding gradients during the training process. We choose the number of vehicles because it has a straightforward relationship with the current traffic load on a particular lane, which can be easily channeled into decisions concerning the traffic lights.

3.2. Action space

In the formulated SUMO infrastructure, there are four lanes where the traffic load can arrive from, and in the intersection, the vehicles are only allowed to turn right or go straight out from the network. For this environment, the utilized discrete actions are the choice of simplicity. There are only two discrete actions. The first one turns the traffic signal to green in the horizontal lanes, while the vertical lanes are red, called North-South Green (NSG). The second does the opposite, notably turns the horizontal lane's traffic light to red and the vertical's to green, called East-West Green (EWG). It is also important to mention that the chosen traffic signal setting stays intact for 5s measured in the SUMO's infrastructure time. Only after that can the agent choose another one.

3.3. Reward scheme

Along with the state representation, the formulation of the reward scheme is another crucial aspect since this scalar feedback value is the only guidance that an agent has in the endeavor of developing an optimal behavior. Moreover, the nature of the rewards is identical to heuristics. Thus they come with no guarantees. This paper introduces a new rewarding for the TSC problem, which arises from our new interpretation of the composed feature-based value vector containing the normalized number of vehicles per lane. The state vector is interpreted as a distribution. The actions can control the distribution's mean and standard deviation by changing the traffic lights and easing the lanes' load. The agent's goal is chosen to minimize the standard deviation of this distribution. For achieving this goal, the reward is immediate. Its value is the distribution's calculated standard deviation, min-max normalized to the [1,-1] interval. The scheme is complemented with conditions that always have to be fulfilled during the training. If not, the episode terminates, and the agent receives a punishment instead of a reward. The conditions determine a threshold regarding the number of vehicles in each lane. If it is exceeded, the episode terminates. The episode is also terminated if the agent does not provide a green phase to a direction that has lower occupancy than the introduced threshold, but the lanes of the other direction are empty. The traffic-wise interpretation of this concept is that the agent must learn that if there is a lane asymmetrically loaded with traffic, it should provide more green time in that particular direction. It is also essential that the agent's way of interfering is easing up heavily loaded lanes, which lowers the number of vehicles that stay on the particular lane. Our expectation from the introduced state-action-reward structure that it can be easily generalized; hence it lets the agent quickly adapt to unseen traffic scenarios.

Of course, this concept operates over the assumption that each lane is identically important, and the traffic load has to be eased equally.

The calculation of the introduced reward is shown in the 4 equation.

$$R = R_{max} + \frac{(\sigma - \sigma_{min} * (R_{min} - R_{max}))}{(\sigma_{max} - \sigma_{min})} \tag{4}$$

The parameters in the reward function are chosen according to the σ calculated from the state vector. If $\sigma < 0.1$ then the parameters are $R_{max} = 1$, $R_{min} = 0$, $\sigma_{max} = 0.1$, $\sigma_{min} = 0$. If $\sigma >= 0.1$ then the parameters are $R_{max} = 0$, $R_{min} = -1$, $\sigma_{max} = 0.5$, $\sigma_{min} = 0.1$.

3.4. Limitations

The study focuses on the TSC problem for the case of a single intersection where the left turns are omitted, and the possible signal phases can not conflict. Thus the agent's only goal is to find the proper phase combination for the given traffic flow.

3.5. Training

The paper aims to formulate a new state representation complemented with a unique reward scheme that enables the high-level generalization of the TSC problem. The most common approach for exploiting the Neural Networks generalization feature is a randomized environment where the agent interacts with many different scenarios and a tremendous number of training episodes, which makes the reservations against RL's sample inefficiency legitimate. Following this concept can result in a solution with excellent performance. Still, we can not measure generalization directly since it is not known which scenario the agent is unfamiliar with. Consequently, the opposite strategy is utilized where the agent interacts with the same scenario in every episode. Thanks to that, we can measure the level of generalization, specifically by evaluating the agent's performance in truly unseen scenarios. The chosen training scenario that the agent tries to solve during the entire training process is the control of an equally loaded intersection, where each lane toward the intersection is assigned with the same amount of traffic flow. The parameters of the training and the Neural Network are shown in Table 1.

Table 1. The parameters of the training.

Parameter	Value
Layers	Dense
Num. of hidden layers	3
Num. of neurons	256,256
Hidden layers activation function	RELU
Learning rate (α)	0.0001
Discount factor (γ)	0.9
Update frequency (ξ) in eps.	10
Optimizer	Adam
kernel initializer	Xavier normal

Figure 2 shows the convergence of the PG algorithm on the introduced scenario. Both the average and the min reward of the episodes are displayed. As mentioned above, the reward is calculated based on the standard deviation of the lane's density. Hence each value can be directly associated with the quality of the control performed by the PG agent.

4. Results

After training the agent, its performance is evaluated in various unseen scenarios that can genuinely assess generalization quality. But first, the agent's behavior is considered in the very traffic scenario used during training. This scenario is when all lanes have equal traffic flow, and the traffic signals have to be controlled accordingly. For this traffic scenario, the solution is obvious; the agent must provide equal green time for both the horizontal and vertical directions, and it does just that. At first sight, it shows that the agent understands its goal and reaches it, but for now, it can be overfitting. Consequently, edge cases are defined to obtain a clear and thorough assessment of the agent's behavior. The first class of cases is the ones where either horizontal or vertical lanes get traffic flow. It goes without

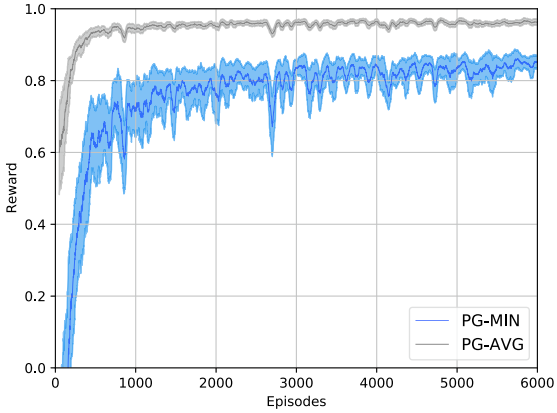

Fig. 2. Convergence of the PG algorithm

saying that the agent has never seen a scenario like that. Nevertheless, it solves the scenario properly by only providing green time to the direction -whether horizontal or vertical- that has traffic flow and keeps the opposite direction's traffic lights red. This behavior shows that the agent understands the control task and the actions' influence on the infrastructure. For that matter, this behavior performs better for sure than a traffic light control with fixed periods in terms of queue length, which shows that the RL-based approach can operate in a traffic-responsive manner. The second class of test cases is where each lane gets traffic flow, but one lane gets considerably more. In these cases, the agent alters the symmetric policy that provides equal green time for all directions. The alteration is substantial when the number of vehicles exceeds the limit, used as a terminal condition in the training procedure. In this scenario, the agent seems to understand what an asymmetric traffic flow means for the intersection, and it acts accordingly. These evaluations suggest that despite the training setup, the abstraction of the control problem along with the reward scheme that arises from the new interpretation of state representation and the simplified discrete actions create a structure that profoundly enhances the generalization of the problem. The trained agent and the SUMO's built-in Time-gap based controller are also compared in the same scenario. Table 2 shows the results. The results show that the agent applies a control that results in fewer vehicles in the most occupied lanes during the comparison in both different scenarios.

Table 2. Comparison of RL agent and Time-Gap based Actuated Control

Maximum density of a link during simulation [%]	RL-Agent	Time-Gap based Actuated Control
Scenario-I	70%	76%
Scenario-II	32%	42%

5. Conclusion

This paper proposes a new abstraction for the TSC problem: a combination of state representation, a brand new rewarding concept, and simplified action space. Together these components boost the generalization of the control problem. As the evaluations underlined, the trained RL agent adapts its behavior to unseen traffic scenarios to operate in the right fashion. In the future, the proposed approach, further versatilities have to be explored. Notably, it is possible to establish a hierarchy in importance between the individual lanes by weighing them to calculate the standard deviation. This consideration can turn the agent more sensible for the traffic load of specific lanes. Moreover, such an approach can manage more interconnected intersections since the complementation would not change the underlying idea.

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